ADITYA DEY

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PROFILE

PhD researcher in long-term time series forecasting (LTSF), designing forecasting architectures with minimal parameters so that they run faster at inference (fewer FLOPs) and are small enough for TinyML deployment. Combines low-rank approximation (LoRA, VeRA, Tied-LoRA, complex-valued LoRA), knowledge distillation from large foundation and supervised models, and signal processing. Brings nine years of IT and data consulting and an MSc in Data Science.

EDUCATION

[Mon YYYY] – present **PhD, [programme]** [University]
Thesis: *"Long Term Time Series Forecasting with Minimal Parameters for Higher Inference"*
Designs LTSF architectures with fewer parameters to lower FLOPs and inference time and to shrink model size to the TinyML range.

Aug 2021 – May 2023 **Master of Science in Data Science** NMBU, Norway
Thesis: feature engineering of numerical and embedded features for prediction with non-linear regression models (Random Forest, SVR, AdaBoost, XGBoost, ANN); hyperparameter search, cross-validation, and model explanation with PDP and SHAP.
Python with OOP and SOLID principles, tested with pytest and tox.

RESEARCH EXPERTISE

- **Parameter-efficient models:** low-rank approximation of weights; LoRA, VeRA, Tied-LoRA and complex-valued LoRA used as standalone forecasting models.
- **Knowledge distillation (KD):** transfer from large foundation or supervised models to lightweight, deployable students (online, offline and self-distillation).
- **Time series analysis:** state-space equations, autocorrelation and cross-correlation, decomposition, seasonality extraction with circulant matrices.
- **Signal processing:** Discrete Fourier Transform (DFT), Discrete Cosine Transform (DCT), Discrete Wavelet Transform (DWT).
- **Architectures:** RNN, LSTM, GRU, Echo State Network (ESN), TCN, MLP, self-attention, patch-based attention (PatchTST, iTransformer).
- **Efficiency targets:** parameter count, FLOPs, inference latency and model size for TinyML and edge deployment.
- **Battery modelling:** P2D lithium-ion models (DFN, SPMe, SPM) and surrogate modelling.

PUBLICATIONS

- [Full citations with Aditya Dey in bold; see Google Scholar profile above]

CONSULTING EXPERIENCE

Sep 2019 – Aug 2021 **Data Analyst** Mastercard, India
Built data pipelines combining BigQuery, Oracle SQL and PySpark; modelled star and SCD schemas.
Planned and delivered Power BI reports with VPs and application leads, including regression analyses; validated data and reports monthly.

Sep 2018 – Sep 2019 **Application Lead** Cognizant, India

Ran Master Data Management reporting across multiple databases and maintained Informatica ETL pipelines with SQL and Python.

Led Python, Informatica and Oracle upgrades and migrations, and optimised existing queries to cut redundancy.

Aug 2012 – Sep 2018 **Application Lead** LTIMindtree, India

Managed production, UAT and business sign-offs, and coordinated releases through CAB meetings.

Resolved defects in shell and Python scripts and Oracle PL/SQL, using SQL tracing to locate faulty procedures and triggers.

LANGUAGE SKILLS

- **English:** [level]
- **Norwegian:** [level]

IT-SKILLS

- **Programming and ML:** Python, PyTorch, TensorFlow, Keras, scikit-learn, SciPy, NumPy
- **Data:** Oracle SQL, MS SQL, PySpark, BigQuery, MongoDB, Redis, Power BI, Matplotlib, Seaborn, Plotly
- **Engineering:** Git, Bitbucket, Jenkins, Ansible, AWS, GCP, pytest, Agile Scrum
- **Writing:** LaTeX

REFERENCES

- Available upon request.